

Integrated Electric Power/Water Distribution System Modeling and Control Under Extreme Mega Drought Scenarios

Scott Zuloaga^{ib}, *Student Member, IEEE*, and Vijay Vittal^{ib}, *Fellow, IEEE*

Abstract—This work considers the operation of the electric power system (EPS) in coordination with the water distribution system (WDS) during extreme operating condition scenarios. The interdependencies are captured by modeling the main dependency of the WDS on the EPS, power for WDS pumps, as well as the main dependency of the EPS on the WDS, the need for cooling water for conventional thermoelectric generation, in simulation. The extreme scenarios include studying the effect of droughts (water shortages) in the WDS and electrical power outages of WDS pumping stations in the EPS. Appropriate network models and mathematical formulations for optimizing control of the operation in the power system with consideration of the present operating conditions in the WDS are presented. These include a novel optimal power flow model for unit dispatch, a model for systematically derating thermoelectric generation during periods of droughts as well as a method for adjusting the cooling water demands of that generation within the WDS. The implementation of these policies is seen to increase the duration for which conventional thermoelectric generation is available for dispatch during such extreme conditions.

Index Terms—Combined economic and environmental dispatch, electric power system, water distribution system, extended time period simulations, system control, water-energy nexus.

NOMENCLATURE

Parameters:

b_l :	Primitive susceptance for line l .
$C_{\text{water},g}$:	Water cost for generator g .
$CR_{\text{water},g}$:	Water consumption rate for generator g .
$C_{g,i}$:	Fuel Cost of segment i for generator g .

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The authors are with the School of Electrical, Computer, and Energy Engineering, Arizona State University, Tempe, AZ 85281 USA (e-mail: szuloaga@asu.edu; vijay.vittal@asu.edu).

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δ_{fuelCost} :	Objective function weight for fuel cost.
$\delta_{\text{waterCost}}$:	Objective function weight for water cost.
EF:	Elasticity of supply factor for the price of water.
$P_{\text{load},nt}$:	Active power demand, bus n , period t .
$P_{\text{pload},nt}$:	Active WDS pump power, bus n , time-period t .
$P_{g,\text{max}}$:	Maximum active power output for generator g .
$P_{g,\text{min}}$:	Minimum active power output for generator g .
$P_{l,\text{max}}$:	Maximum active power flow through element l .
$P_{l,\text{min}}$:	Minimum active power flow through element l .
$Q_{t,g}$:	Cooling water supplied by WDS to generator g , time-period t .
$R_{g,\text{hr}}$:	Hourly ramp rate for generator g .
RSU_g :	Startup ramp rate for generator g .
RSD_g :	Shutdown ramp rate for generator g .
$TC_{\text{tankRefill}}$:	Time constant for WDS _{DA} components.
TPF :	Percentage that onsite water storage tank level can change over present time-step.
Υ_{gt} :	Fixed on/off status of generator g , time-period t , output from SCUC solution and input to CEED.
WCR _g :	Water consumption rate (gal/MWH), generator g .
W_{derate} :	Weight for TPF in derating LP.

Variables:

$cROC_t$:	Cumulative rate of change of averaged onsite storage tank level, time-period t .
P_{lt} :	Active power flow on line l , time period t .
θ_{nt} :	Voltage angle of bus n , time period t .
$P_{g/g_{th},t}$:	Active power output for generator g/g_{th} , time-period t .
P_{gti} :	Active power output for segment i , generator g , time-period t .
$P_{\text{max},i,g}^*$:	Post-processing variable for derating LP
$P_{\text{max},i,g}$:	New maximum active power output for generator g , time-period i .
ROC_t :	Averaged onsite storage tank rate of change.
$st_{g,t}$:	Slack variable for time-period t , generator g .
TL_{gt} :	Water tank level for generator g , time period t .
$WDS_{\text{DA},i}$:	Demand adjustment factor for altering node demands of generators within WDS simulation, $i \in \{1, 2, \text{Tot}\}$.

I. INTRODUCTION TO THE WATER-ENERGY NEXUS

MODERN infrastructure systems have grown increasingly complex in order to meet the needs of our technologically advanced society. The authors in [1] highlight critical infrastructure systems in the US and the important role that these systems play in society. Although these systems are complex in their own right, dependence of some systems on others and even interdependence between the systems exists. A high-level overview of the field of critical interdependent infrastructure systems is given by the authors in [2], and the term *interdependency* is defined as “a bidirectional relationship between two infrastructures through which the state of each infrastructure influences or is correlated to the state of the other.”

The electric power system (EPS) and the water distribution system (WDS) have several interdependencies that have been recognized and which are referred to as the water-energy nexus. These interdependencies arise because power generation via thermoelectric plants requires water for the plants’ cooling cycle and because water taken from its source requires electricity for pumping and/or treatment needs before its final delivery to the end user. The energy requirements for the WDS can be a large portion of the total energy requirements of a city or state, as evidenced in California in 2005 when the state required 19% of its total energy use for water system requirements [3]. In 2012, it was estimated that 12.3% of the total energy consumption in the US was for water related uses [4]. Likewise, water usage in the EPS can constitute a large percentage of the total water demands, being up to as much as 45% of the total daily water withdrawn from water sources in the U.S. [5].

It is also seen from the data in [5] that the usage of water by the power production sector in each state varies quite widely, with values of over 80% of daily water withdrawals in Alabama, Illinois, and S. Carolina down to much smaller values including approximately only 1.71% for thermoelectric power generation in Arizona. Although water for thermoelectric generation may represent a small amount of the overall demand in the WDS, the converse is not necessarily true. Fig. 1 shows that the average percentage of total energy production by thermoelectric generation in Arizona from 1990 through 2018 is a little over 90%. Thus, this serves as an example of the importance of having adequate supplies of cooling water to maintain grid integrity.

Given that the two systems are dependent on each other in normal operation as described above, it is reasonable to assume that the need to study the interaction of the two systems and to develop models with which to perform these studies will be essential to maintaining satisfactory operation of both systems in the future. This is a consequence of the fact that increased demand or even unforeseen events will cause the two critical infrastructure systems to be more heavily burdened.

Some examples of increased stress or failure of one system due to the other will now be given. A long-term outage of a water treatment facility in Tampa, Florida [6] due to an electrical failure resulted in accompanying health concerns due to a lack of pressure within the underground water system pipes which normally keeps pollutants from entering and contaminating the water supply. Natural events can also disrupt normal operation of

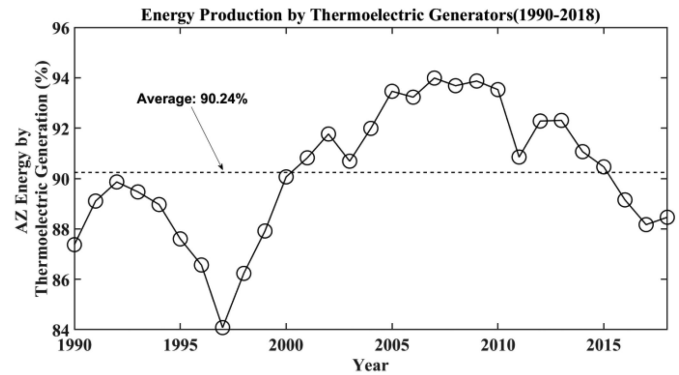


Fig. 1. AZ energy production by thermoelectric generation 1990–2018 [36].

either system, such as when drought conditions in the southeast [7] resulted in the need to reduce certain nuclear generation by almost 50%. Massive flooding in Iowa resulted in substation outages which affected the ability of WDS pumping stations to prevent water from entering and contaminating a water treatment facility [8]. Other evidence of previous disruptions can be seen in places such as Howard County, MD, where electrical outages to two distribution feeders after Hurricane Sandy caused an overflow at the Little Patuxent sewage plant and the spillage of almost 20 million gallons of untreated effluent [9]. This prompted the building of onsite generation which can completely supply the treatment plant’s electrical needs [10]. The South Monmouth Regional Sewerage Authority provides an example [11] of preemptive action taken to safeguard against such a catastrophic failure as the sewerage authority built a mobile trailer to house the plant’s primary controls and electric equipment which allows it to be moved to safety and temporarily replaced with a cheaper set of electronics during severe storms. Finally, it is noted that further motivation to study the interactions of these two systems in greater detail is provided by the fact that there are extreme events, especially in the American Southwest, which could arise in the near future given the increased probability of extreme mega-drought risks [12].

II. MODELING INFORMATION

Although water is used for several processes at thermoelectric generation plants (e.g., the steam cycle) [13], the water consumption of these processes is comparatively less than the water that is used for the cooling cycle which is estimated to be around 90% of the total water usage [14]. Therefore, the primary dependency of the EPS on the WDS is the withdrawal and/or consumption of water for the cooling cycle [15]. The water losses associated with a wet, closed-loop cooling cycle are from drift, blowdown and evaporation. Drift occurs as water droplets are blown out of the cooling tower due to the air flow, blowdown is the required removal of water to meet water quality requirements as the cooling water is recirculated and evaporation is the process of water changing phases from liquid to vapor [7]. Thermoelectric generation stations in the area of interest (American Southwest) are normally supplied by two independent water sources for redundancy. Additionally, generation sites normally have a finite

amount of water storage onsite which can supply the needs for the water intensive processes of the generation units for an extended period of time (weeks). From historical data kept by utilities on the annual amount of water consumed at various plants as well as a plant's total yearly energy production, water consumption rates (gal/MW·Hr) can be calculated for use in relating a plant's active power output to its water consumption.

For the WDS, the primary dependency on the EPS is electrical power for WDS pumps. Similar to the information compiled by electric utilities, water utilities keep information on anticipated water demands, system topology, as well as pump specifications which can be used with operational quantities (pressure head and flow rates) to calculate power consumption. Operational control decisions in the WDS are normally made on an hourly basis and this is reflected in standard WDS time-domain simulation time-step durations from which desired pump operation decisions are determined.

Given this description of the behavior of the key dependencies of the two systems, it is clear that representation of the power system at the small time-scales required for representing the fast electrical and electromechanical phenomena is not necessary as changes in onsite water storage (EPS) and pump switching decisions (WDS) happen over relatively much longer time-periods. Time-series power flows are therefore used to represent the EPS over the duration of the simulation. Along with these static AC power flows, the operational control decisions in the EPS are accomplished by using DC models of the network within a short-term unit commitment model along with an optimal power flow (OPF) [16] model for determining unit dispatch in coordination with values for cooling water delivery from the WDS and the WDS pump power demands.

The OPF which is presented later is a modified version of traditional OPF formulations and falls under what is sometimes called a combined economic and environmental dispatch (CEED). The authors in [17] summarize modified versions of the OPF which consider environmental aspects of generating electric power, such as emissions. While previous work includes environmental concerns in the form of emissions [17], [18], it has not been until recently that constraints relating to generator's use of water have begun to garner interest. The authors in [19] examine the effects on thermoelectric generation dispatch for units that have once-through cooling and that are physically located on the same waterway. Water constraints are now also being considered within planning studies as evidenced by the Western Electricity Coordinating Council's (WECC) Long-Term Planning Tool that has integrated water supply constraints into the transmission expansion planning problem by including the amount of available water in the watershed of where a generation unit is located [20].

Network solutions for the WDS are found by performing a hydraulic analysis of the WDS which results in knowledge of pressures at nodes in the system, flows in the links, water quality concentrations at nodes, and the water levels within tanks. The optimization of the WDS was accomplished by interfacing a control optimization model for determining an optimal pump schedule with the hydraulic analysis.

A more detailed description of the unit commitment model implemented for this study as well as the full description of the WDS network equations, optimization model and case studies can be found in [21]–[23]. Recent work which has looked at co-optimization schemes for the WDS and EPS are presented in [24]–[27], where the emphasis on interdependency is primarily one-way and captured through the inclusion of WDS pump loads in the EPS. An optimal control methodology for the coordinated operation of the WDS and EPS is presented in [24]. This control methodology minimized operational costs for both systems but it did not consider or include WDS real-time operation or emergency conditions. In [25], the power system is represented via a distribution system only and the only coupling between the two systems is through the power consumed by the pumps but the issue of pump scheduling is not addressed and the simulations considered are short-term. Given the availability of storage at thermoelectric power plants and in the water towers in the WDS, it is essential to consider long-term simulations in order to examine the impact of disruptions. The authors in [26] and [27] represent the power system only at the distribution level and again the interdependency of the two systems is solely due to power consumed by WDS pumps. In summary, [25]–[27] model the EPS at the distribution system level only and those references along with [24] have no consideration of the consequences of WDS operations on thermoelectric generation. In addition, these works do not include anything other than normal system conditions in the simulations conducted, which is in contrast to one of the main motivations behind this work, which aims to consider the operation of the WDS and EPS under extreme conditions.

With those recent works in mind, the key contributions of this paper are as follows:

- 1) Long-term simulations which capture both the inherent, slow interdependent EPS and WDS quantity changes as well as some of the slower control decisions that are made in either system
- 2) Explicit representation of the main dependencies of each system on the other (cooling water for thermoelectric generation and power for WDS pumps)
- 3) A dispatch formulation that explicitly considers system conditions with respect to the WDS
- 4) A systematic approach to de-rate thermoelectric plants during droughts based on the current levels in the on-site water storage tanks at the plant location

Lastly, it is noted that the co-optimization formulations in [24]–[27] seek to minimize the total cost or amount of water used in the WDS and/or minimize the energy consumption/costs of the WDS pumps in addition to minimizing power system operation costs. In contrast, the extensive formulation for the WDS optimization used in this work seeks to minimize the difference between water requested at WDS nodes and the amount which is supplied, subject to network and operational constraints as detailed in [21]. This serves as another distinction between the presented work and existing studies and it is noted that the different assumptions and simplifications for a co-optimization formulation would likewise yield different results.

As information sharing between entities in the EPS with those in the WDS is virtually non-existent, the approach to modeling, simulation and control presented in this work offers a reasonable attempt at optimizing operations of the two systems with limited data exchange.

III. MATHEMATICAL FORMULATIONS

The mathematical formulation for the modified OPF in the form of a CEED as well as the mathematical formulations for the three implemented EPS control policies will now be presented. The availability of information from the WDS allows for new decisions to be made in the EPS simulation with regards to the operation of the system. The objective function of the CEED consists of a weighted sum of the fuel costs as well as the operational costs associated with water. Given that there is an inverse relationship between the incremental fuel cost for a thermoelectric unit and its water consumption rate, this gives rise to a trade-off needing to be made between system cost and water consumption. This allows for the dispatch of generation to be influenced by selection of weights for these terms and by which this formulation differs from other efforts. The first of the three control policies seeks to adjust the maximum output of generators experiencing extreme conditions to ensure that the units' power output is available for a longer period of time by reducing their water consumption. Since severe drought conditions can result in water demands not being met given the lack of water availability, this policy seeks to offer a systematic approach to implement curtailment of a unit's maximum power output during such extreme drought conditions as this is one of the most common actions that is taken. The second policy is a more immediate measure and is an adjustment of the operational cost of water based on the recent supply of cooling water demands. During such conditions where water requirements are not being supplied, a power plant has to use more water from its onsite storage, so the storage levels have to be carefully controlled and the altering of the dispatch (lowering dispatch and hence lowering total water consumption) along these lines is addressed in this policy. The third policy affects the nodal demand of generator cooling water requirements which are seen within the WDS. This offers a way to exert increased influence on the WDS optimization procedure to help ensure a higher fulfillment of power plant cooling water make-up demands throughout the course of the simulation. Given that there is no systematic approach to effectively control all the three interdependent aspects mentioned above, the control policies presented in this work are formulated to address these aspects in an integrated fashion.

A. CEED Formulation

The optimization model for the dispatch of generation using information from the WDS determines the real power set-points for all units which have been committed for the next t time-periods following the solution of the unit commitment problem.

Minimize

$$\sum_t \left(\frac{\delta_{\text{fuelCost}}}{\delta_{\text{waterCost}}} \sum_g \sum_i (C_{gi} P_{gti}) + (C_{R_{\text{water},g}} C_{\text{water},g} \sum_i P_{gti}) \right) \quad (1)$$

$$\text{subject to } P_{gt1} = P_{g,\min} \Upsilon_{gt} \quad (2)$$

$$0 \leq P_{gti} \leq P_{gi,\max} \Upsilon_{gt} \quad (3)$$

$$P_{gt} = \sum_i (P_{gti}) \quad (4)$$

$$P_{lt} - b_l (\theta_{n(to),t} - \theta_{n(from),t}) = 0 \quad (5)$$

$$P_{l,\min} \leq P_{lt} \leq P_{l,\max} \quad (6)$$

$$\theta_{rt} = 0 \quad (7)$$

$$\left(\frac{\sum_{G \in G_n} P_{gt} + \sum_{\forall l(to)} P_{lt} - \sum_{\forall l(from)} P_{lt} - (P_{\text{load},nt} + P_{\text{pload},nt})}{\sum_{\forall l(from)} P_{lt} - (P_{\text{load},nt} + P_{\text{pload},nt})} \right) = 0 \quad (8)$$

$$P_{g,\min} \Upsilon_{gt} \leq P_{gt} \leq P_{g,\max} \Upsilon_{gt} \quad (9)$$

The modified CEED objective function is the weighted sum of two groups of cost functions, with the decision variables being $P_{gt/gth,i}$ and the subscript gth denoting thermoelectric generation. The first group captures the cost related to fuel consumption. The fuel cost curves are assumed here to be piece-wise linear functions of the unit's output with values of the i -th piece-wise segment's cost equal to C_{gi} . The second group of terms in the objective function gives an operational cost to the amount of cooling water that is consumed. The incremental cost of water (C_{water}), in \$/gal, is variable and the adjustment of this value is described in the Control Policy II Section (III.C) below. The term $CR_{\text{water},g}$ is a unit's consumption rate of water in gal/MW·Hr. This can be calculated from historical data or given a variable value based on known conditions such as ambient temperature and air conditions at the generation site if this level of accuracy is desired and such information is known. Since the fuel cost of a given unit is usually inversely related to its water consumption rate, the two cost terms are in direct competition with each other and therefore neither can be minimized, except for the extreme case of large weights, δ_{fuelCost} or $\delta_{\text{waterCost}}$. The summation for both terms is over all time periods (e.g., 24-hours) and over all generators for the fuel cost term and over the subset of all thermal units for the terms constituting operational water costs. A short discussion of the effect of the weights follows, where it is assumed that δ_{fuelCost} has a value between 0 and 1 and $\delta_{\text{waterCost}}$ is simply 1 minus δ_{fuelCost} . With δ_{fuelCost} having a value of 1, there is no environmental consideration since the terms related to the operational water costs equal zero and the CEED formulation becomes that of a normal optimal power flow problem. Conversely, with δ_{fuelCost} equal to 0, the CEED becomes what can be thought of as a purely "environmental" dispatch. Thus, the effect of increasing δ_{fuelCost} is to move the dispatch away from cheap (water cost) towards cheap (fuel cost) generation.

Equation (2) ensures that the first segment of the piece-wise real power output curve is zero if the unit commitment on/off variable is zero. Constraint (3) gives the maximum power outputs of each piece-wise segment while (4) sets the active power output of the unit equal to the sum of the individual segments of the piece-wise linear curve. Equation (5) is the linear approximation to power flow across a transmission element, where the active power flow magnitude is equal to the difference between the to bus voltage angle and the from bus angle multiplied by the element's primitive susceptance. Constraint (6) limits the line

flows to be between a line's minimum and maximum values. Equation (7) states that the voltage angle of the reference node, r , is zero. Equation (8) is the node active power supply-demand balance equation. The equation states, going over terms in words from left to right, that the sum of all generation at a bus plus the sum of all active power leaving through transmission elements at that bus is equal to the active power coming into the bus through terminating transmission elements plus the active power demand at the bus. Note that the demand is specified explicitly in terms of the electric system load at the bus plus any WDS pump load ($P_{\text{load},nt}$) present at the bus. Lastly, constraint (9) ensures that a unit's output is between its minimum and maximum output values when the unit is on and zero when the unit is off, dictated by the on/off variable input from the unit commitment solution, Υ_{gt} .

B. Control Policy I: Unit Derating

This section presents a method for curtailing a unit's output under extreme conditions. The number of high impact, weather related electric power grid disturbance has been growing, and [28] describes how just under 50% of large blackouts in the US have been caused by severe weather. Given that the majority of non-regulatory stay actions for thermoelectric generation experiencing cooling water issues have been the curtailment of generation [29], a formal method for determining the maximum output of such units was desired. The mathematical formulation of the curtailment model is as follows:

$$\text{Minimize } - \sum_{t,g} P_{t,g} + \sum_{t,g} 1x10^{10} s_{t,g} \quad (10)$$

$$\text{subject to } P_{t,g} \leq P_{g,\max} \Upsilon_{gt} \quad (11)$$

$$(P_{t,g} - P_{t-1,g})(1 - \text{abs}(\Upsilon_{g,t-1} - \Upsilon_{gt})) \leq R_{g,hr} \quad (12)$$

$$(P_{t-1,g} - P_{t,g})(1 - \text{abs}(\Upsilon_{g,t-1} - \Upsilon_{gt})) \leq R_{g,hr} \quad (13)$$

$$(P_{t,g} - P_{t-1,g})(1 - \Upsilon_{g,t-1}) \leq RSU_g \quad (14)$$

$$(P_{t-1,g} - P_{t,g})(1 - \Upsilon_{gt}) \leq RSD_g \quad (15)$$

$$\sum_t P_{t,g} \cdot WCR_g \cdot T - s_{t,g} \leq \sum_t Q_{t,g} \cdot T + TPF \cdot TL_g \quad (16)$$

$$s_{t,g} \geq 0 \quad (17)$$

The objective function (10) seeks to maximize the $P_{t,g}$ values, where the variable $P_{t,g}$ is used to avoid confusion with the nominal maximum output, $P_{g,\max}$. The unit commitment variables in (11) a value is found only for the units which are committed in each time period. Constraints (12) and (13) seek to keep the fluctuation of $P_{g,\max}$ within the running ramp rates of the units. Similarly, constraints (14) and (15) handle the ramp rates if the unit is committed or turned off during one of the time periods of this model's implementation. Constraint (16) utilizes input data from the WDS simulation. In words, the left-hand side is the algebraic sum of the product of the decision variable and the unit's water consumption rate and a slack variable to ensure a feasible solution of the LP. The right-hand side is the sum of all the water flows into the generator storage tank from the WDS plus an additional factor which is discussed below. So,

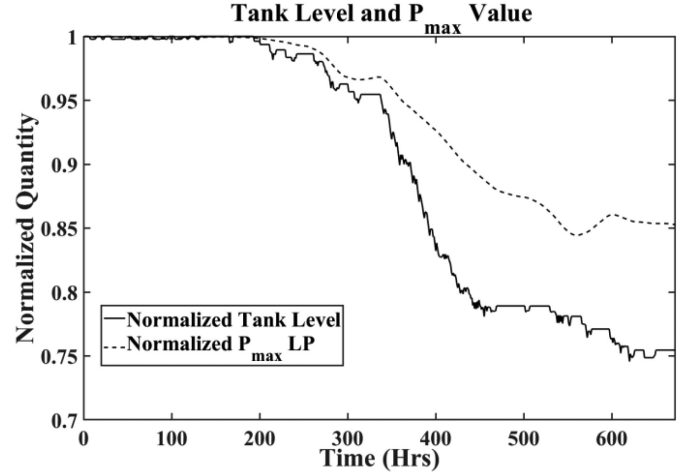


Fig. 2. Normalized onsite water storage tank level and maximum active power.

this constraint seeks to keep the water consumed less than the water supplied, plus an additional quantity determined by the present onsite water storage tank level. Lastly, (17) keeps the slack variables positive so that they are minimized as shown in the objective function. The right-most term in (16) is the product of the tank percentage factor and the tank level at the present iteration of the simulation defined mathematically as:

$$TPF = W_{\text{derate}} \cdot (1 - TL_g / TL_{g,\text{initial}}) \quad (18)$$

where $TL_{g,\text{initial}}$ is the initial onsite storage tank level, TL_g is the tank level in the present iteration and W_{derate} is a weight determined from sensitivity analysis. The output of the LP model requires additional post-processing which is done through the averaging of two sets of data as follows. First, the results for the t time-periods within the LP are averaged giving a single value, $P_{\max,i,g}^*$, for each unit. This is then followed by the averaging of the previous t simulation time-period $P_{g,\max}$ values in order to reduce any large fluctuations in this value that could occur due to correspondingly large fluctuations in $Q_{t,g}$ values which again are the water flow rates into the onsite storage tanks nodes and come from the WDS optimization/network solution. Mathematically, this is accomplished by:

$$P_{\max,i,g}^* = \text{ave}_{\text{non-zero}}(P_{1:t,g}) \quad (19)$$

$$P_{\max,i,g} = \text{ave}(P_{\max,i,t-1:g} + P_{\max,i,g}^*) \quad (20)$$

Additionally, the weight in (18), W_{derate} , was determined through sensitivity analysis and a value of 0.02 was used in all of the following simulations and results. The choice of this value results in the behavior of a $P_{g,\max}$ value as is shown in Fig. 2. This figure shows plots of normalized tank level, which the unit is de-rated based upon, along with the normalized $P_{g,\max}$ values for a comparison of how the derating scheme follows the level of onsite water storage. As can be seen, the derating scheme exhibits the desired behavior (tracking the tank level as it changes) and the lag between the storage tank level dropping and when the $P_{g,\max}$ value starts to decrease is due to the averaging that is done in the post-processing.

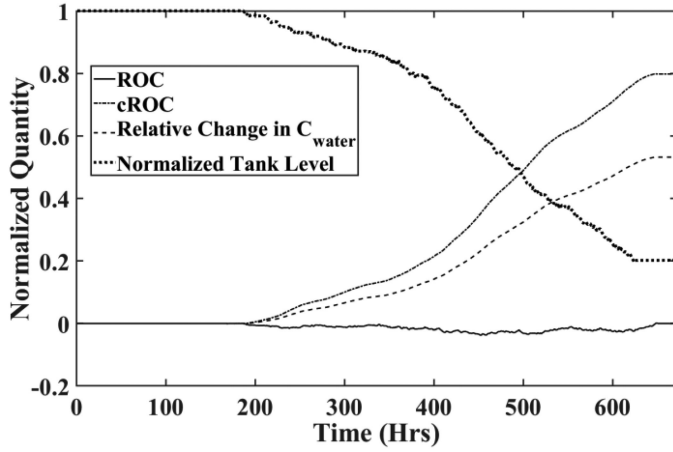


Fig. 3. Implementation of operational water cost adjustment.

C. Control Policy II: Operational Water Cost Modification

In extreme drought scenarios, it might also be desirable to further influence the dispatch of generation in the affected geographical area in order to keep units operational for as long as possible by limiting the use of onsite water storage. The approach presented here is the adjustment of the operational water cost in (1) based on how the onsite cooling water demands are being met. Defining a tank level rate of change as:

$$ROC_t = ave_{t-i:t}(s(t)) \quad (21)$$

where the quantity s is defined as the difference between the cooling water which was supplied by the WDS and that which was actually requested (consumed) by a unit in the EPS.

$$s(t) = \frac{Q_{t,g,\text{supplied}} - Q_{t,g,\text{requested}}}{P_{g,\max 0} \cdot WCR_g} \quad (22)$$

A cumulative rate of change is then defined as:

$$cROC_t = cROC_{t-1} - ROC_t \quad (23)$$

And finally, the operational cost of water, which is analogous to the incremental fuel cost, is calculated in (24) using an assumed elasticity of supply factor, EF.

$$C_{\text{water},g} = C_{\text{water},g,\text{initial}} \cdot (1 + cROC_t \cdot EF) \quad (24)$$

The price elasticity of supply is the ratio of a change in quantity over a change in price. The inverse of this value gives a change in price for a change in quantity and thus this value can be used in relating the change in quantity of onsite water storage level to a price for adjusting the operational water cost. Fig. 3 shows the behavior of the ROC_t , $cROC_t$ and $C_{\text{water},g}$ values for a scenario during which the water storage tank level underwent the changes as shown.

D. Control Policy III: WDS Demand Adjustment

Since the implemented WDS optimization adjusts the demands which are supplied based on the time of day and other constraints, the cooling water make-up requirements which are node demands within the WDS simulation are adjusted in

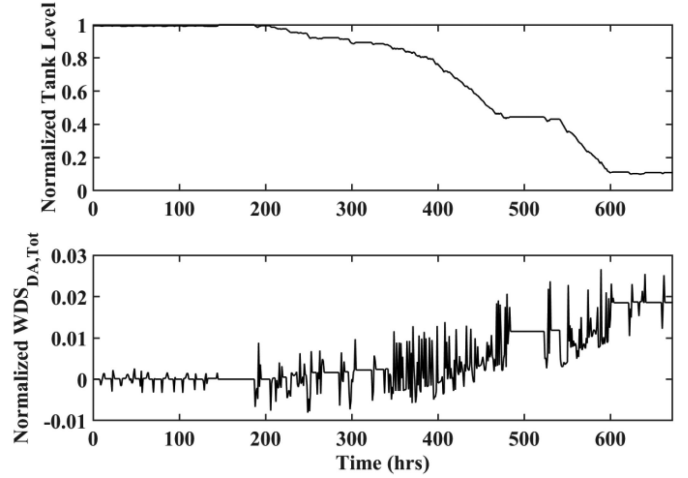


Fig. 4. Onsite storage tank level and WDS demand adjustment factor.

order to help refill onsite storage tank levels after they have dropped. The node demands which are used as input for the WDS simulation therefore consist of both the actual cooling water requirements for the given simulation time-period as well a demand adjustment component. This demand adjustment factor has two elements. The first element, $WDS_{DA,1}$ in (25), takes into account whether the make-up requirements were satisfied in the previous simulation time-period. The second component, $WDS_{DA,2}$ in (26), aims to bring the tank level back to its original value.

$$WDS_{DA,1} = \frac{Q_{t,g,\text{supplied}} - Q_{t,g,\text{requested}}}{TC_{\text{tankRefill}}} \quad (25)$$

$$WDS_{DA,2} = \left(1 - \frac{\text{mod}(TL_g, TL_{g,\text{initial}})}{TL_{g,\text{initial}}}\right) \frac{TL_{g,\text{initial}}}{TC_{\text{tankRefill}}} \quad (26)$$

where mod is the modulus operator. The demand adjustment factor is then calculated as the algebraic sum of (25) and (26):

$$WDS_{DA,Tot} = WDS_{DA,1} + WDS_{DA,2} \quad (27)$$

Fig. 4 shows the plots of the normalized tank level as well as the demand adjustment factor, on different ordinate scales. The main takeaway from the lower figure that plots the total demand adjustment factor, $WDS_{DA,Tot}$ (27), is that there are two clearly visible components. The first is a gradual, slow increase in the demand adjustment factor. This corresponds to the corrective action because of (26) and as demonstrated above. The high frequency component in the figure is the result of (25), which captures the variations from time-period to time-period resulting from the thermoelectric cooling water make-up requirements either being met or not.

IV. SIMULATION METHODOLOGY

The coupling between the water system and the power system was achieved through alternatively exchanging data between the power system and water network simulations via a program control overlay developed in Python. Fig. 5 shows this concept

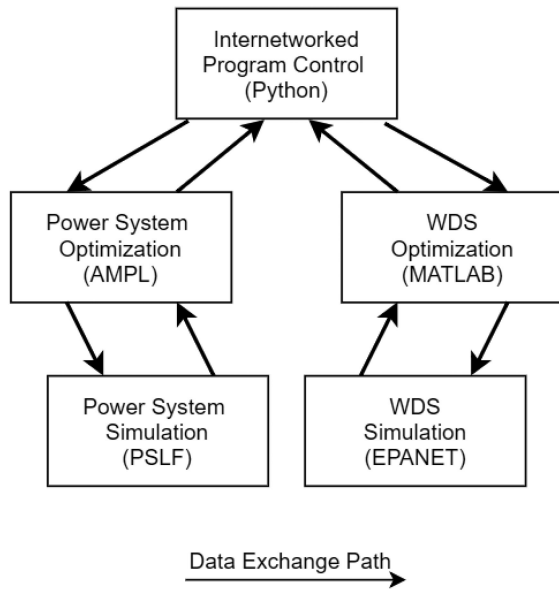


Fig. 5. Simulation engine overview.

at a high level. The relevant simulation data exchanged between the power system and water network simulations included EPS network configurations, pump power consumption for inclusion in power system loads, and thermal generation cooling water requirements that were used as node demands within the water system. The temporal dependence of one system on the other is accomplished through the modification of loads within the power system and the modification of demands within the water system. The spatial relationship between the two systems is accounted for by placing those loads or demands at the appropriate location within the EPS and WDS. All the optimization models for the optimization and control of the power system were solved via implementation within the AMPL programming language [30] using Gurobi [31] as a solver. The complete EPS network solution which obtained the net active and reactive power injections as well as the voltage magnitudes and angles at each bus was obtained using GE's PSLF software [32]. More details of this simulation engine including a program flow chart can be found in [21].

V. EXAMPLE APPLICATION

A. Test System

The test system for the EPS which was used in simulation was a modified version of the IEEE 14 bus system [33], which is shown in Fig. 6. The modifications included the addition of representative distribution systems [34] to which the water system pumping station loads were mapped since WDS pumping stations are connected to the electric power distribution system. This network model therefore includes both levels of the system (transmission and distribution) in order to account for the fact that WDS pumps are in the distribution system while still taking into consideration the thermoelectric generation that is in the bulk power system. Data for the distribution systems was generated from the IEEE 34-bus test distribution system [35].

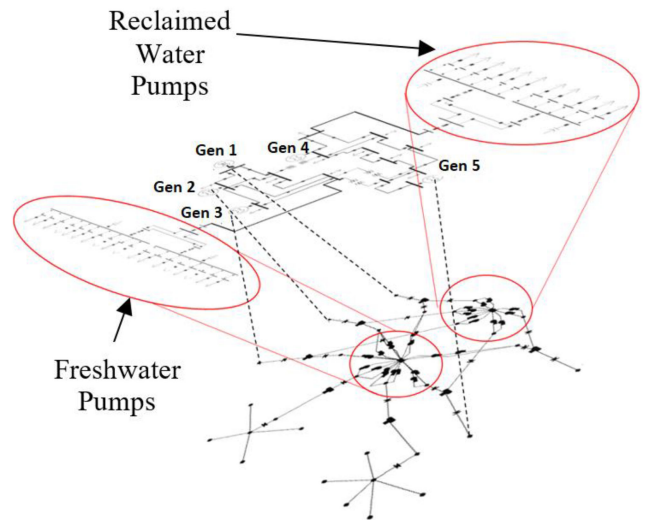


Fig. 6. WDS and EPS test systems and spatial mapping between them.

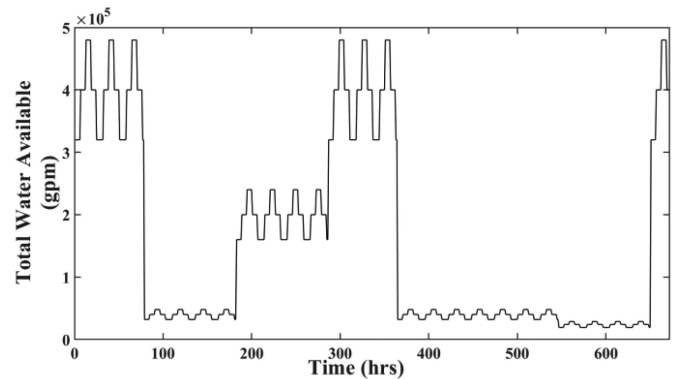


Fig. 7. Total WDS water availability.

Also shown in Fig. 6 is the mapping of generators in the EPS to nodes in the WDS. The data for this test system including network parameters, assigned generator fuel-types, historical water consumption data, and a complete description and the data for the WDS test system is given [21].

B. Application Scenarios

The results presented are a case study which offers a comparison of the direct effect that the different policy implementations have on overall EPS performance during a scenario with emulation of severe drought conditions and disruptions to the supply of electricity to pumping stations. The scenario setup includes the total system water availability as shown in Fig. 7. With a total system mean water demand of around 150,000 gpm, the simulation is clearly divided into periods which represent more water than needed (beginning and end), shortages of water emulating drought (end of week 1, week 3, start of week 4) and a period during which demands should be able to be satisfied (week 2) within the WDS. In addition, long term power outages affecting the WDS pumping stations represent the addition of more extreme system conditions. The pumps

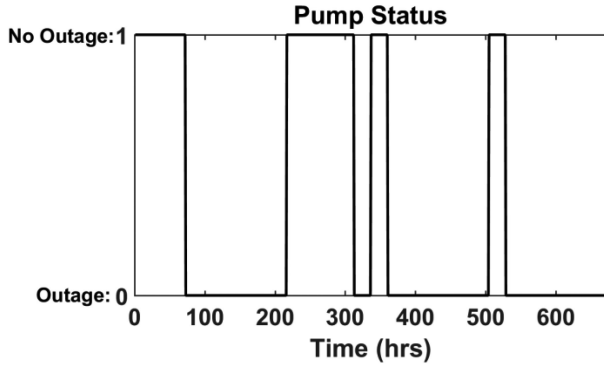


Fig. 8. Pump outage occurrence.

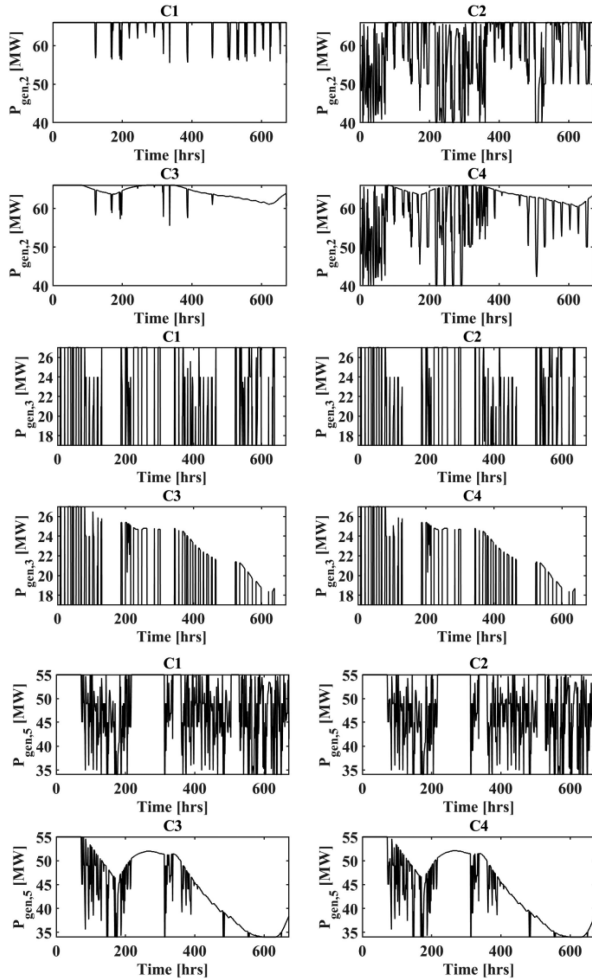


Fig. 9. Active power outputs(left-to-right): Unit 2 (Npp), Unit 3 (CT), and Unit 5 (CCGT).

experiencing an outage can be seen in [21] and the hours during which outages occurred is shown in Fig. 8. Control Policy III (WDS demand pattern adjustment) and equal values of δ_{fuelCost} and $\delta_{\text{waterCost}}$ were used in the following cases. Likewise, the WDS simulation-optimization methodology in [21] was used, including the time-dependent weights for WDS demand pattern satisfaction which seeks to serve non-residential WDS demands in off-peak hours. The cases examined were:

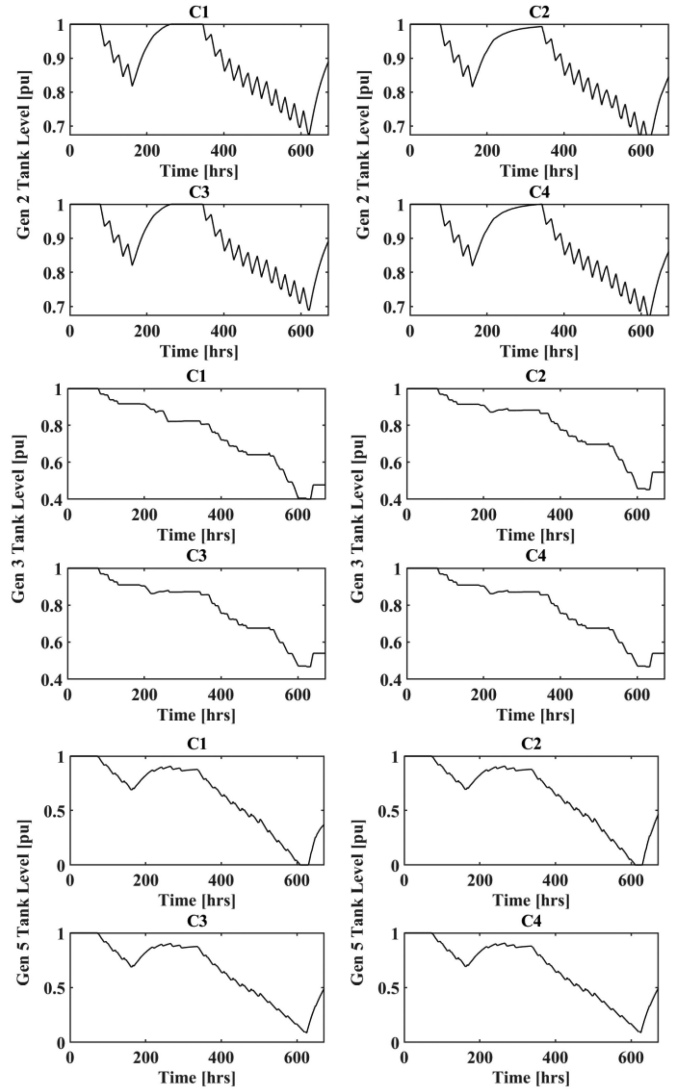


Fig. 10. Tank levels (left-to-right): Unit 2 (Npp), Unit 3 (CT), and Unit 5 (CCGT).

Case 1 (C1): CEED only

Case 2 (C2): CEED with Policy II: Operational Water Cost Adjustment

Case 3 (C3): CEED with Policy I: Unit Derating, LP Model

Case 4 (C4): CEED with Policy I and Policy II

VI. RESULTS DISCUSSION

Fig. 9 shows the active power outputs for Unit 2, nuclear power plant (NPP), Unit 3, combustion turbine (CT), and Unit 5, combined cycle gas turbine (CCGT). Subplots are shown for each unit corresponding to each of the cases (C1-C4) above. It is readily apparent that the inclusion of the operational water cost adjustment causes much more variation in the dispatch of the baseload unit (Unit 2) since the cost to dispatch becomes relatively more expensive as compared with the other units that have lower water consumption rates. The effect of unit derating policy is also evident for all units in the bottom two subplots

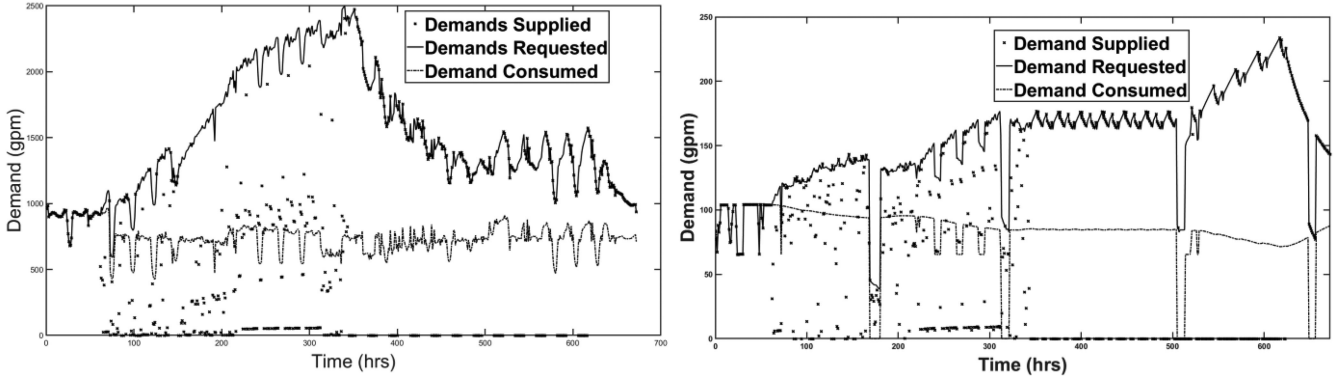


Fig. 11. Pump outage and water shortage effect on make-up cooling water supplied and effect of policy III for Units 2 (L) and Unit 5 (R).

in each figure as the reduction in the $P_{\max}$ values following decreasing onsite storage takes effect.

Fig. 10 shows the levels of the onsite water storage in each case for the same generation units. The policy implementations are seen to result in improvements by increasing the minimum value in the storage tank levels as well as increasing the amount of water onsite at the end of the simulation, and this is particularly evident for Units 3-5. The effect of Policy III is clear at the end of the simulation as onsite storage levels begin to rise as the WDS has enough water to meet demands within the system. The effect of the WDS demand pattern weights is also evident in the tank levels as the lower priority peaking units have much larger decreases in their storage tank levels and the non-monotonically decrease for Unit 2 demonstrates priority given to the base load units over WDS residential demands in off-peak hours.

The effect of including pumping station outages in the scenario definition on the control policies implemented occurs as follows. Pump outages affect the pump scheduling done in the WDS optimization procedure as there are fewer pumps available to use and a harder task of meeting demands while satisfying network and operations constraints in the WDS. As a result, the node demands in the WDS for the generators (cooling water make-up requirements) might not be supplied in a given time-period in simulation. The likelihood of this is increased with less water available system-wide. Fig. 11 shows the make-up cooling water demands, the cooling water supplied by the WDS, and the amount requested by Units 2 and 5 for a scenario similar to C4. Here, the effect of cooling water not supplied results in increasing node demands for the two units as a result of Policy III. It is noted that this increase has two visible components. The slower overall increase in demand requested is a result of $WDS_{DA,2}$ while the high frequency changes are due to $WDS_{DA,1}$ as make-up demands are (or are not) being met in a given time-period. The result of water consumed being greater than what is supplied then effects the operational water cost adjustment in Policy II and the long-term effect of consumption being greater than the supply is a reduction in the onsite water storage tank level as seen in Fig. 10. This reduction in onsite water storage tank levels along with the consumed cooling water not being supplied directly affects the output of Policy I.

TABLE I
GENERATOR AVERAGE ACTIVE POWER COMPARISON, C1-C4

	GENERATOR	C1 (MW)	C2 (MW)	C3 (MW)	C4 (MW)
2	Npp	65.5	58.9	64.2	59.8
3	CT	24.0	23.9	23.2	23.2
5	CCGT	48.8	48.6	44.5	44.4

TABLE II
GENERATOR MINIMUM TANK LEVEL COMPARISON, C1-C4

	GENERATOR	C1 (PU)	C2 (PU)	C3 (PU)	C4 (PU)
2	Npp	0.6734	0.6439	0.6880	0.6637
3	CT	0.3989	0.4509	0.4663	0.4660
5	CCGT	0.0000	0.0000	0.0849	0.0875

TABLE III
GENERATOR END OF SIMULATION TANK LEVEL COMPARISON, C1-C4

	GENERATOR	C1 (PU)	C2 (PU)	C3 (PU)	C4 (PU)
2	Npp	0.890	0.846	0.892	0.861
3	CT	0.477	0.545	0.539	0.539
5	CCGT	0.372	0.466	0.491	0.494

TABLE IV
GENERATOR AVERAGE ACTIVE POWER COMPARISON, C2

	GENERATOR	$\Delta_{FUEL\text{COST}}=0.0$	$\Delta_{FUEL\text{COST}}=0.5$	$\Delta_{FUEL\text{COST}}=1.0$
2	Npp	58.9	64.9	65.6
5	CCGT	48.7	48.8	45.6

Tables I–III summarize relevant data from the simulations. Table I indicates how the base load unit dispatch can be lowered and therefore kept operational longer (by not consuming as much cooling water) during the drought periods. The improvement in system performance for the cases in comparison to the base case (C1) by means of increased minimum tank levels and higher storage levels at the end of the simulation for Units 3 and 5 is also noted.

Lastly, it is noted that the weight applied to the terms associated with water costs ($\delta_{\text{waterCost}}$) as well as the actual values of the operational cost of water ($C_{\text{water},g}$) can have a substantial impact on the resulting dispatch of generation. Table IV illustrates this

point by showing the active power for Units 2 and 5 for the C2 scenario and varying $\delta_{\text{waterCost}}$ values and it is noted that the CEED dispatch moves from generation that is cheap (fuel) to generation that is cheap (water) and thus the average output of Unit 2 decreases approximately 10.2% while Unit 5 increases approximately 6.8% as the δ_{fuelCost} weight decreases.

VII. CONCLUSION

Background on the water-energy nexus and motivations for the type of network and control modeling for extended period simulations has been given. The analytical framework for the control schemes that were presented have been detailed and implemented within a complex simulation procedure for operational time-scale simulations. Simulations of the coordinated operation of the WDS and the EPS have been conducted and a case study featuring results from several simulations were shown. The developed simulation engine and its associated modeling and control implementations would allow an operation planner to conduct analysis of the type normally done by a water utility (pump scheduling) and a utility or ISO/RTO (Unit Commitment and Unit Dispatch) while allowing for the additional benefits derived from having knowledge of the infrastructure system that each is dependent on. The dispatch formulation (CEED) explicitly considers system conditions with respect to the WDS via inclusion of a unit's operational water cost (reflecting onsite water storage tank levels) and inclusion of WDS pump loads. The operational water cost is adjusted by application of one of the presented control policies depending on the severity of the drought conditions being experienced. Another of the policies provides an explicit and systematic approach to derate thermoelectric plants during droughts based on the present levels in the on-site water storage tanks at the plant location. It was seen that implementation of the control policies improved system performance with respect to the interdependent system quantities by discouraging the high dispatch of base load units (reducing water consumption) and raising minimum and final values of onsite water storage tank levels.

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REFERENCES

- [1] President's Commission on Critical Infrastructure Protection, *Critical Foundations: Protecting America's Infrastructures*. Washington, DC, USA, Oct. 1997.
- [2] S. M. Rinaldi, J. P. Peerenboom, and T. K. Kelly, "Identifying, understanding, and analyzing critical infrastructure interdependencies," *IEEE Control Syst. Mag.*, vol. 21, no. 6, pp. 11–25, Dec. 2001.
- [3] California Energy Commission, Integrated Energy Policy Report, Sacramento, CA, USA, Rep. CEC-100-2005-007CMF, 2005.
- [4] S. C. Jones and R. B. Sowby, "Quantifying energy use in the U.S. public water industry — a summary," *ASCE EWRI Currents*, vol. 16, no. 4, pp. 6–9, 2014.
- [5] M. A. Maupin *et al.*, "Estimated use of water in the United States in 2010: U.S. geological survey circular 1405," US DOI/USGS, Reston, VA, USA, 2014.
- [6] R. Danielson, "Treatment plant power outage means all of tampa must boil water," Tampa Bay Times, 2013. [Online]. Available: <https://www.tampabay.com/news/localgovernment/boil-water-notice-issued-for-all-tampa-water-customers/1276215>
- [7] K. Gerdes and C. Nichols, "Water requirements for existing and emerging thermoelectric plant technologies," NETL, Golden, CO, USA, Rep. 402/080108, 2009.
- [8] Admin, "Lessons from 1993 guide flood fight now," Radio Iowa, 2008. [Online]. Available: <https://www.radioiowa.com/2008/06/10/lessons-from-1993-guide-flood-fight-now/>
- [9] A. Metcalf, "Millions of gallons of raw sewage overflows into little patuxent," Patch, 2012. [Online]. Available: <https://patch.com/maryland/columbia/report-millions-of-gallons-of-raw-sewage-overflows-in8e907dc022>
- [10] A. Yeage, "On eve of Sandy anniversary, Ulman breaks ground on sewage plant backup generators," The Baltimore Sun, 2013. [Online]. Available: <https://www.baltimoresun.com/news/maryland/howard/lisbon-fulton/ph-sewage-power-fail-safe-story.html>
- [11] U.S. EPA, "Power resilience: Guide for water and wastewater utilities," U.S. EPA Office of Water, Rep. EPA 800-R-19-001, 2019.
- [12] B. I. Cook, T. R. Ault, and J. E. Smerdon, "Unprecedented 21st century drought risk in the American Southwest and Central Plains," *Sci. Adv.*, vol. 1, no. 1, pp. 1–8, 2015.
- [13] J. Maulbetsch and B. Barker, "Water use for electric power generation," Electric Power Research Institute, Palo Alto, CA, Rep. 1014026, 2008.
- [14] L. Badr, G. Boardman, and J. Bigger, "Review of water use in U.S. thermoelectric power plants," *J. Energy Eng.*, vol. 138, no. 4, pp. 246–257, Mar. 2012.
- [15] T. Feeley *et al.*, "Addressing the critical link between fossil energy and water," The DOE/Office of Fossil Energy's Water Related Research, Development, and Demonstration Programs/NETL, Washington, DC, USA, 2005.
- [16] B. Stott and J. L. Marinho, "Linear programming for power-system network security applications," *IEEE Trans. Power Appar. Syst.*, vol. PAS-98, no. 3, pp. 837–848, May/Jun. 1979.
- [17] J. H. Talaq, F. El-Hawary, and M. E. El-Hawary, "A Summary of environmental/economic dispatch algorithms," *IEEE Trans. Power Syst.*, vol. 9, no. 3, pp. 1508–1516, Aug. 1994.
- [18] R. Yokoyama, S. H. Bae, T. Morita, and H. Sasaki, "Multiobjective optimal generation dispatch based on probability security criteria," *IEEE Trans. Power Syst.*, vol. 3, no. 1, pp. 317–324, Feb. 1988.
- [19] D. Fooladivanda and J. A. Taylor, "Dispatching thermal power plants under water constraints," in *Proc. 53rd Annu. Allerton Conf.*, Sep./Oct., 2015, pp. 396–401.
- [20] V. C. Tidwell *et al.*, "Water supply as a constraint on transmission expansion planning in the Western interconnection," *Environ. Res. Lett.*, vol. 11, no. 12, pp. 1–15, Nov. 2016.
- [21] S. Zuloaga, P. Khatavkar, L. W. Mays, and V. Vittal, "Interdependent electric and water infrastructure modeling, optimization and control," *IET Energy Syst. Integr. J.*, vol. 2, no. 1, pp. 9–21, Mar. 2020.
- [22] P. Khatavkar and L. W. Mays, "Testing an optimization/simulation model for the real-time operations of water distribution systems under limited power availability," in *Proc. Congr. Tech. Adv.: Infrastructure Resilience Energy - Papers Sessions 1st Congr. Techn. Adv.*, 2017, pp. 1–9.
- [23] P. Khatavkar and L. W. Mays, "Model for real-time operations of water distribution systems under limited electrical power availability with consideration of water quality," *J. Water Resour. Plan. Manag.*, vol. 144, no. 11, pp. 1–8, 2018.
- [24] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 1101–1110, Jan. 2019.
- [25] A. S. Zamzam, E. Dall'Anese, C. Zhao, J. A. Taylor, and N. D. Sidiropoulos, "Optimal water-power flow-problem: Formulation and distributed optimal solution," *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 1, pp. 37–47, Mar. 2019.
- [26] Q. Li, S. Yu, A. Al-Sumaiti, and K. Turitsyn, "Micro water-energy nexus: Optimal demand-side management and quasi-convex hull relaxation," *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 4, pp. 1313–1322, Dec. 2019.
- [27] Q. Li, S. Yu, A. Al-Sumaiti, and K. Turitsyn, "Modeling and co-optimization of a micro water-energy nexus for smart communities," in *Proc. Innovative Smart Grid Technol. Conf. Eur. (ISGT-Eur.)*, 2018, pp. 1–5.
- [28] R. J. Campbell, "Weather-related power outages and electric system resiliency," Washington, DC, USA, Rep. R42696, 2012.
- [29] J. McCall, J. Macknick, and D. Hillman, "Water-related power plant curtailments: An overview of incidents and contributing factors," NREL, Golden, CO, USA, Rep. TP-6A20-67084, 2016.

- [30] R. Fourer, D. M. Gay, and B. W. Kernighan, "A modeling language for mathematical programming," *Manage. Sci.*, vol. 36, no. 5, pp. 519–554, 1990.
- [31] Z. Gu, E. Rothberg, and R. Bixby, "Gurobi optimizer reference manual, version 5.1," 2013.
- [32] S. Barnes, R. D'Aquila, and B. Thomas, "PSLF user manual," General Electric (GE), Schenectady, NY, USA, 2011.
- [33] Illinois Center for a Smarter Electric Grid, "IEEE 14-bus system," 2018. [Online]. Available: <http://icseg.iti.illinois.edu/ieee-14-bus-system>
- [34] Schneider Electric, "Recommended electrical network design for efficient plant and energy operations," Schneider Electric, Rueil-Malmaison, France, 2012.
- [35] K. P. Schneider *et al.*, "Analytic considerations and design basis for the IEEE distribution test feeders," *IEEE Trans. Power Syst.*, vol. 33, no. 3, pp. 3181–3188, May, 2018.
- [36] Net Generation by State by Type of Producer by Energy Source; US EIA, 2019. [Online]. Available: <https://www.eia.gov/electricity/data/state/>

Scott Zuloaga (Student Member, IEEE) received the B.S.E. degree in electrical engineering and the Ph.D. degree from Arizona State University, Tempe, AZ, USA, in 2015 and May 2020, respectively. He recently joined the dynamic studies group as a Planning Engineer with the Electric Reliability Council of Texas.

Vijay Vittal (Fellow, IEEE) received the B.E. degree in electrical engineering from the B.M.S. College of Engineering, Bangalore, India, in 1977, the M. Tech. degree from the Indian Institute of Technology, Kanpur, India, in 1979, and the Ph.D. degree from Iowa State University, Ames, IA, USA, in 1982. He is a Regents. Professor and the Ira A. Fulton Chair Professor with the Department of Electrical, Computer, and Energy Engineering, Arizona State University, Tempe, AZ, USA. He is currently the Director of the Power System Engineering Research Center (PSERC). He is a member of the National Academy of Engineering.